

Milestone 01

Answer Key

Environment

Update the YAML with your first and last name.

Load the packages from the code-along. (Install first if not already done so.)

```
library(tidyverse)
library(gssr)
library(gssrdoc)
library(haven)
library(summarytools)
```

Load all years of the GSS data.

Then, use the `gss_get_yr()` function to load the data only for the 2024 survey respondents. (*Hint:* Use the code from code-along-01.qmd)

```
# Load the data (will appear in your Global Environment pane)
data(gss_all)

# load the gss 2024 data (ADD YOUR CODE BELOW)
gss24 <- gss_get_yr(2024)
```

Fetching: https://gss.norc.umd.edu/documents/stata/2024_stata.zip

Code, Output, Meaning

Use R to complete the checkpoints below. Show your work (e.g., R code chunks) where appropriate. Add narrative (text outside code chunks) or comments (text inside the code chunks) throughout.

Reference specific statistics (where appropriate) from your output to justify your answers. Explain what the values tell you about the data; interpret their meaning in relation to the question.

Use the `gss_all` dataset for checkpoints 01 - 04.

Use the `gss24` dataset for checkpoints 05 - 10.

Checkpoint 01

Report the survey question that corresponds with the `fefam` variable. How did you find out?

?fefam

Survey question: Now I'm going to read several more statements. As I read each one, please tell me whether you strongly agree, agree, disagree, or strongly disagree with it. d. It is much better for everyone involved if the man is the achiever outside the home and the woman takes care of the home and family.

Checkpoint 02

Is `fefam` a dichotomous, nominal, ordinal, or interval-ratio variable? Justify your answer.

`fefam` is an ordinal variable because the response categories (e.g., Strongly Agree, Agree, etc.) have a meaningful order but there are not equal intervals (such as numeric values).

Checkpoint 03

Use `gss_which_years()` below to show the survey years with the variable `fefam` in the `gss_all` dataset.

```
gss_which_years(gss_all, fefam)
```

```
# A tibble: 35 x 2
  year    fefam
  <dbl> <lgl>
1 1972    FALSE
2 1973    FALSE
3 1974    FALSE
4 1975    FALSE
5 1976    FALSE
6 1977     TRUE
7 1978    FALSE
8 1980    FALSE
9 1982    FALSE
10 1983    FALSE
# i 25 more rows
```

Which years of the survey did the fefam variable appear in the GSS?

1977, 1985-1986, 1988 - 1991, 1993-1994, 1996-2018 (every other year), 2021-2022, 2024

Checkpoint 04

Use `print_labels()` to see a list of the values and labels for the `fefam` variable.

```
print_labels(gss_all$fefam)
```

```
Labels:
value          label
1          strongly agree
2              agree
3          disagree
4    strongly disagree
NA(d)         don't know
NA(i)              iap
NA(j)    I don't have a job
NA(m)          dk, na, iap
NA(n)          no answer
NA(p)        not imputable
NA(r)          refused
NA(s)    skipped on web
NA(u)          uncodeable
```

NA(x) not available in this release
NA(y) not available in this year
NA(z) see codebook

If a respondent has a value of 4 for the fefam variable, what does that mean?

The respondent strongly disagreed with the statement “It is much better for everyone involved if the man is the achiever outside the home and the woman takes care of the home and family.”

Checkpoint 05

Convert the `fefam` variable to a factor variable.

1. Use `zap_missing()` to get rid of all the ‘missing’ (NA) levels.
2. Use `as_factor()` to replace the values with labels.
3. Use `drop_levels()` to get rid of the empty levels (if any).

```
gss24$fefam <- zap_missing(gss24$fefam)
gss24$fefam <- as_factor(gss24$fefam)
gss24$fefam <- drop_levels(gss24$fefam)
```

Checkpoint 06

Use `table()` to show the frequencies of the `fefam` variable.

```
table(gss24$fefam)
```

strongly agree	agree	disagree	strongly disagree
167	492	899	604

Checkpoint 07

Use `freq()` to create a (pretty) relative frequency table to show for the `fefam` variable using the 2024 data. Set the following function options:

- `report.nas = FALSE`
- `headings = FALSE`

```
freq(gss24$fefam, report.nas = FALSE, headings = FALSE)
```

	Freq	%	% Cum.
strongly agree	167	7.72	7.72
agree	492	22.76	30.48
disagree	899	41.58	72.06
strongly disagree	604	27.94	100.00
Total	2162	100.00	100.00

How many respondents “strongly agree” with the fefam statement in 2024? What % of respondents either “strongly agree” or “agree”?

167 respondents strongly agree with the statement. 30.48% of respondents either agreed or strongly agreed with the statement.

Checkpoint 08

Use `freq()` to create a (pretty) relative frequency table for the `happy` variable using the 2024 data.

```
freq(gss24$happy, report.nas = FALSE, headings = FALSE)
```

Tagged NA values were detected and will be reported as regular NA; use `haven::as_factor()` to

	Freq	%	% Cum.
very happy [1]	684	20.85	20.85
pretty happy [2]	1892	57.67	78.51
not too happy [3]	705	21.49	100.00
Total	3281	100.00	100.00

Were most respondents “not too happy,” “pretty happy,” or “very happy” in 2024?

Most respondents (57.67%) were pretty happy in 2024.

Checkpoint 09

Create a (pretty) cross-tab of the **happy** and **sex** variables using only the 2024 survey data. Make sure your values are labelled for easy interpretation.

Use `zap_missing()`, `as_factor()`, `droplevels()`, and `ctable()` as appropriate. Set the following function options for `ctable()`:

- `prop = "c"`
- `format = "p"`
- `useNA = "no"`

```
# Fix the sex variable
gss24$sex <- zap_missing(gss24$sex)
gss24$sex <- as_factor(gss24$sex)
gss24$sex <- droplevels(gss24$sex)

# Fix the happy variable
gss24$happy <- zap_missing(gss24$happy)
gss24$happy <- as_factor(gss24$happy)
gss24$happy <- droplevels(gss24$happy)

# Create a cross-tab
ctable(gss24$happy, gss24$sex,
  prop = "c",
  format = "p",
  useNA = "no"
)
```

Cross-Tabulation, Column Proportions

happy * sex

Data Frame: gss24

	sex	male	female	Total
happy				
very happy		293 (20.2%)	386 (21.3%)	679 (20.8%)
pretty happy		830 (57.1%)	1054 (58.2%)	1884 (57.7%)
not too happy		330 (22.7%)	370 (20.4%)	700 (21.5%)
Total		1453 (100.0%)	1810 (100.0%)	3263 (100.0%)

Were women or men more likely to report they were very happy?

Women (21.3%) were slightly more likely than men (20.2%) to say they were very happy.

Checkpoint 10

According to the table above, were a larger percentage of women or men not too happy in 2024?

A slightly larger percentage of men (22.7%) than women (20.4%) were not too happy in 2024.

Meet the IPUMS Data

To keep your Research Brief progress on track, you'll complete short exercises that correspond with the new course material using your own dataset as part of your milestones.

Consider a research topic you are interested in and explore the documentation for some of the available [IPUMS \(microdata\) datasets](#) that could be relevant for the subject. Start broadly, such as looking at the [Topical Availability Summary](#). Then, look at a couple of individual datasets more in depth. Explore the sidebar of available documentation for 2-3 datasets: About, Data, Documentation (including User Guides and Questionnaires, FAQs).

- Write 1-2 sentences about your research interests.
- Then, write 2-3 sentences about which IPUMS dataset you are thinking of using, explaining the sample population and time frame of the dataset, and why it is a good fit for your research interests.
- Describe 2-3 key variables available in the dataset that are a good fit for your research topic. Report the exact survey questions and response options for these variables.
- Finally, create an IPUMS account for the dataset of interest. See this [help video](#) for details.

NOTE: For your Research Brief, you must choose an [IPUMS \(microdata\) dataset](#) to use.